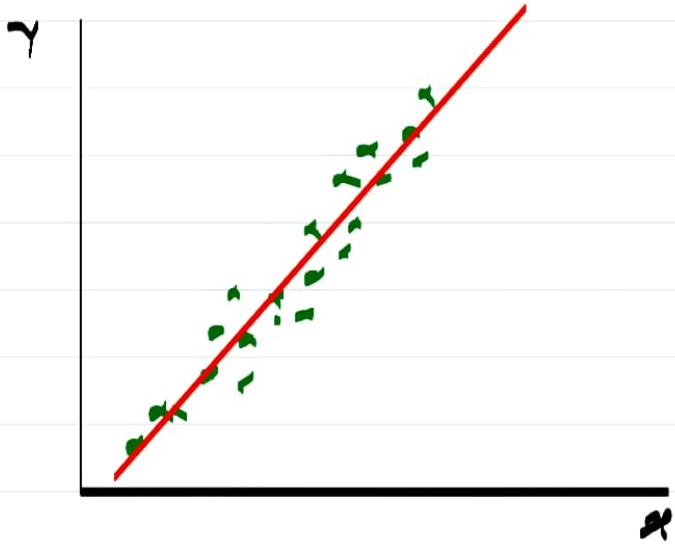
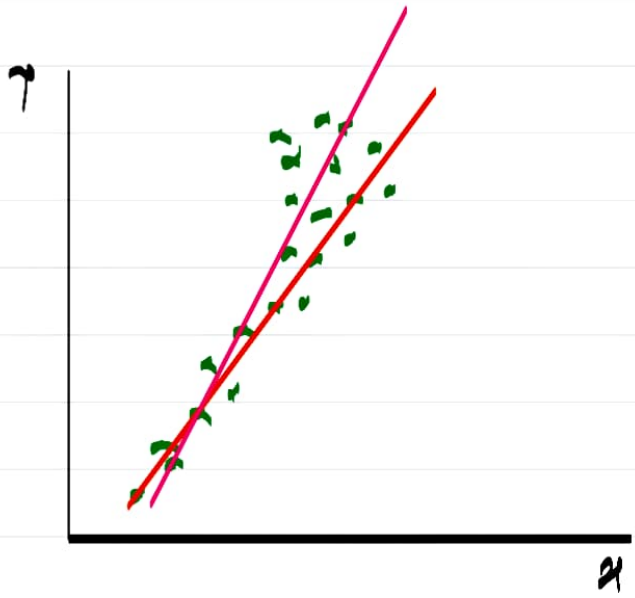
Machine Learning

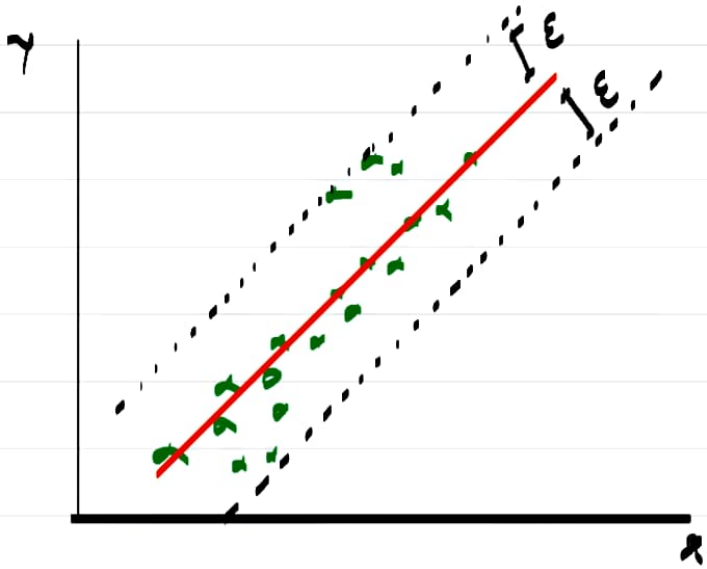
22-05-26

SVM Regression





- the best fit line gets shifted due to outliers in linear regression



lets say a datapoint di has Yact: 4.5 and Ypred = 4

in linear regression

loss: $(Y_{act} - Y_{pred})^2$

$= (4.5 - 4)^2$

$= 0.25$

model got feedback from loss that Ypred needs to be changed to the reduce the loss

this will cause line to shift towards 4.5

In svm regression

$\varepsilon = 1$

loss: $\max[0, |Y_{act} - Y_{pred}| - \varepsilon]$

$= \max[0, |4.5 - 4| - 1]$

$= \max[0, -0.5]$

loss: 0

loss: 0 is a feedback to the model that Ypred doesn't need to shift towards 4.5

the line won't shift in this case as $|Y_{act} - Y_{pred}|$ is within error limits

svm Regression loss function

$$L = \frac{1}{n} \sum_{i=1}^{n} \max[0, |Y_i - \hat{Y}_i| - \varepsilon]$$

ε - hyperparameter for error limit

if $|y_i - \hat{y}_i| < \varepsilon \rightarrow$ loss = 0

if $|y_i - \hat{y}_i| > \varepsilon \rightarrow$ loss = $|y_i - \hat{y}_i| - \varepsilon$

This property makes svm regression less sensitive to outliers compared to linear regression

$$L = \frac{1}{n} \sum_{i=1}^{n} \max [0, |y_i - \hat{y}_i| - \varepsilon]$$

$$L = \frac{1}{n} \sum_{i=1}^{n} \max [0, |y_i - wx_i| - \varepsilon]$$

train the model using gradient descent

find $\frac{\partial L}{\partial w}$

$$w_{new} = w_{old} - \alpha \frac{\partial L}{\partial w_{old}}$$

- Kernel trick will be used for non-linear regression using svm

Regularization:

$$L = C \times \frac{1}{n} \sum_{i=1}^{n} \max [0, |y_i - \hat{y}_i| - \varepsilon] + \sum_{j=1}^{m} w_j^2 \Big/ \sum_{j=1}^{m} |w_j|$$

L2 L1